Erv Wilson's Hexadic Diamond Derived from the A_4 Root System

Abstract

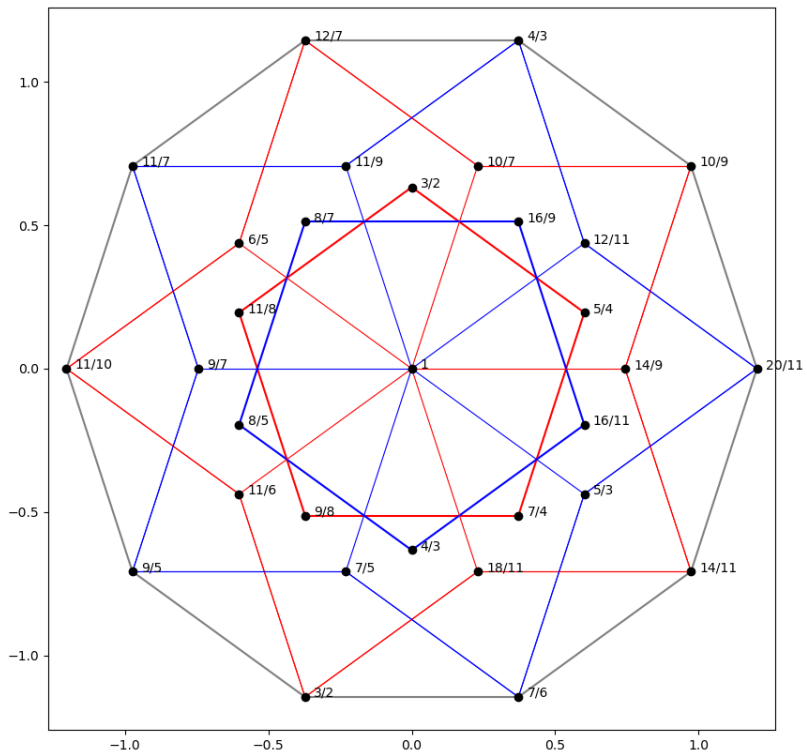
We show that Erv Wilson's Hexadic Diamond—a structure from just-intonation music theory—is fully derivable from the Coxeter theory of the A_4 root system. Every element of the diamond (its 31 points, their projection to the plane, and the connectivity of its star lines) emerges from a single algebraic object: the ambient space of RootSystem(A_4). The only external input is the assignment of odd harmonics $\{3, 5, 7, 9, 11\}$ to the five basis vectors of that space. The same Coxeter-theoretic machinery underlies the cut-and-project construction of Penrose tilings [2, 3].

1 Introduction

The notion that Erv Wilson's Hexadic Diamond may somehow be derived from the projection step of generating a Penrose Tiling came from regarding Figure 2.12 of Senechal. Here, a Voronoi cell – or 5D hypercube – is projected onto a plane orthogonal to $\{1, 1, 1, 1, 1\}$. Though the spacing is off, this figure has all the essential elements of the Hexadic Diamond. The projection of basis vectors in Figure 2.13 of Senechal give a further hint: the project and its reflection render the fundamental otonalities and utonalities of the Hexadic diamond.

Taking a lattice in 5D, and considering axes to be associated with integers 3,5,7,9 and 11, respectively, a connection is made in the usual musical sense (cite Benson). For example, the point (0,-1,1,0,0) corresponds to the ratio 7/5. Taking only the resulting ratios included in the Hexadic Diamond and projecting these points using the projection operator for the // space taken in the usual cut and project procedure for generating a Penrose Tiling gives the placement of tones present in the Hexadic Diamond – albeit with a 90° rotation.

Inspection of the ratios that include both an otonal and utonal element, and the connection between ratios by lines in the Hexadic Diamond hint at some deeper structure. There are 20 of these, and they correspond precisely with the A_4 root system. This can be seen by examining Fig. 3 of reference [3], which shows the The 20 A_4 roots, projected on the Coxeter plane. Connecting lines present in the Hexadic Diamond correspond to orbits of these roots.



2 The A_4 Root System

The root system A_4 has rank 4, with 20 roots living in the hyperplane $H = \{\mathbf{x} \in \mathbb{R}^5 : \sum x_i = 0\}$. The roots are the 20 vectors $\varepsilon_i - \varepsilon_j$ for $i \neq j \in \{0, 1, 2, 3, 4\}$, where $\varepsilon_0, \ldots, \varepsilon_4$ are the standard basis vectors of $\mathbb{R}^5$. The four simple roots are

$$\alpha_1 = \varepsilon_1 - \varepsilon_2, \quad \alpha_2 = \varepsilon_2 - \varepsilon_3, \quad \alpha_3 = \varepsilon_3 - \varepsilon_4, \quad \alpha_4 = \varepsilon_4 - \varepsilon_5,$$

corresponding to the nodes of the A_4 Dynkin diagram ∘–∘–∘–∘. The Cartan matrix encoding inner products between simple roots is

$$C = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

Under the harmonic assignment $\varepsilon_i \mapsto p_i$ with $(p_0, p_1, p_2, p_3, p_4) = (3, 5, 7, 9, 11)$, each root $\varepsilon_i - \varepsilon_j$ maps to the just-intonation ratio p_i/p_j, yielding the 20 combination-product ratios of the Hexadic Diamond (e.g. 3/5, 7/9, 11/3).

3 The Coxeter Element and Projection

3.1 The cyclic permutation matrix

The Coxeter element of $W(A_4) \cong S_5$ is the product of all simple reflections in sequence:

$$c = s_1 s_2 s_3 s_4,$$

where s_i swaps $\varepsilon_i \leftrightarrow \varepsilon_{i+1}$. This product is the cyclic permutation (1 2 3 4 5), represented by the 5×5 circulant matrix

$$P = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

which has order 5 (the Coxeter number $h = 5$). This is the same matrix used in development of the Penrose tiling.

3.2 Eigenspaces over $\mathbb{Q}(\zeta_5)$

The eigenvalues of P are the fifth roots of unity $\omega^k = e^{2\pi i k/5}$. The five eigenspaces decompose $\mathbb{R}^5$ as follows:

- $k = 0$ (eigenvalue 1): the direction $(1, 1, 1, 1, 1)$, orthogonal to H.
- $k = 1, 4$ (eigenvalues $e^{\pm 2\pi i/5}$): the real 2-plane $E_\parallel$, where P acts as rotation by 72. This is the *physical space* of the Penrose tiling.
- $k = 2, 3$ (eigenvalues $e^{\pm 4\pi i/5}$): the real 2-plane $E_\perp$, where P acts as rotation by 144. This is the *internal space* of the Penrose tiling.

The projection of any lattice vector $\mathbf{v} = (v_0, \ldots, v_4)$ to $E_\parallel$ is encoded as

$$z = \sum_{k=0}^{4} v_k \, \zeta^k \;\in\; \mathbb{Q}(\zeta_5),$$

where $\zeta = e^{2\pi i/5}$. The coordinates in the projected plane are $(\mathrm{Re}(z), \mathrm{Im}(z))$.

4 The 31 Points of the Diamond

Count	5D vectors	Coxeter identity	Musical meaning	Projected ring
1	$(0, 0, 0, 0, 0)$	zero vector	unison 1/1	origin
5	$+\varepsilon_i$	standard rep weights	otonal: $3, 5, 7, 9, 11$	inner
5	$-\varepsilon_i$	dual weights	utonal: $1/3, \ldots, 1/11$	inner
10	$\varepsilon_i - \varepsilon_{i+1}$	adjacent roots ($d = 1$)	neighbor ratios	middle
10	$\varepsilon_i - \varepsilon_{i+2}$	non-adjacent roots ($d = 2$)	separated ratios	outer

The inner ring forms a decagon (two interleaved pentagons: otonal and utonal, offset by 36). The outer/middle radius ratio is exactly the golden ratio $\tau = (1 + \sqrt{5})/2$.

5 The Golden Ratio from Cyclotomic Arithmetic

The 20 roots project to two distinct distances depending on the cyclic distance d between their indices:

$$\begin{aligned} d = 1: &\quad |1-\zeta|^2 = 2-\zeta-\zeta^4, \\ d = 2: &\quad |1-\zeta^2|^2 = 2-\zeta^2-\zeta^3. \end{aligned}$$

The ratio of squared norms, computed entirely within $\mathbb{Q}(\zeta_5)$,

$$\tau^2 = \frac{2-\zeta^2-\zeta^3}{2-\zeta-\zeta^4} = \frac{(1+\sqrt{5})^2}{4} \approx 2.618,$$

confirms that the outer/middle distance ratio is τ. This follows directly from $\cos(2\pi/5) = (\sqrt{5}-1)/4$ and $\cos(4\pi/5) = -(\sqrt{5}+1)/4$, both expressions in the golden ratio arising from the eigenvalue structure [2].

6 Star-Line Connectivity via Coxeter Orbits

The star lines of the Hexadic Diamond are generated by the action of P on roots. Because $P \cdot (\varepsilon_i - \varepsilon_j) = \varepsilon_{i+1} - \varepsilon_{j+1}$ (indices mod 5), repeated application of P to any root generates an orbit of five roots. The two star patterns use four such orbits:

- **Red star** — outer orbit of $\varepsilon_0 - \varepsilon_2$ (non-adjacent, $d = 2$); bridge orbit of $\varepsilon_1 - \varepsilon_2$ (adjacent, $d = 1$). This collection shares the ratio $\frac{3}{2}$ with the otonal (red) pentagon.
- **Blue star** — outer orbit of $\varepsilon_2 - \varepsilon_0$; bridge orbit of $\varepsilon_2 - \varepsilon_1$, containing the ratio $\frac{4}{3}$ of the utonal (blue) pentagon.

7 Relationship to Penrose Tilings

The Hexadic Diamond and Penrose tilings share similar Coxeter-theoretic foundations. For the Penrose tiling, it is necessary to consider the Coxeter pair $\{I_2^5, A_4\}$ as detailed in reference [3].

TABLE BELOW NEEDS TO BE CORRECTED WITH CONSIDERATION OF BOYLE & STEINHARDT: THE CONCEPT OF COEXETER PAIRS

	Penrose tiling	Hexadic Diamond
Source	$\mathbb{Z}^5$ lattice	A_4 roots + weights + origin
Selection	geometric window in $E_\perp$	combinatorial (≤ 1 pos, 1 neg exponent)
Projection space	$E_\parallel$ (72 eigenplane)	$E_\parallel$ (same eigenplane)
Result	infinite aperiodic tiling	finite 31-point diagram
Golden ratio	tile length ratio, inflation	outer/middle root ring ratio
Root vectors	tile edge directions	musical interval ratios

Both constructions diagonalize the same permutation matrix P to obtain $E_\parallel$, then project lattice structures to this plane. The Penrose tiling uses the full $\mathbb{Z}^5$ lattice with a geometric (window) filter; the Hexadic Diamond uses a finite, algebraically natural subset determined by the root system itself.

8 The Single External Input

The only element not derived from A_4 is the harmonic assignment:

$$\varepsilon_0 \mapsto 3, \quad \varepsilon_1 \mapsto 5, \quad \varepsilon_2 \mapsto 7, \quad \varepsilon_3 \mapsto 9, \quad \varepsilon_4 \mapsto 11.$$

A lattice vector (a, b, c, d, e) then represents the ratio $3^a \cdot 5^b \cdot 7^c \cdot 9^d \cdot 11^e$. Everything else—the projection geometry, the golden ratio in the radial structure, the star-line connectivity, the otonal/utonal duality—follows from Coxeter theory.

References

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[2] M. Senechal, *Quasicrystals and Geometry*, Cambridge University Press, 1995.

[3] L. Boyle and P. J. Steinhardt, *Coxeter pairs, Ammann patterns, and Penrose-like tilings*, Phys. Rev. B **106**, 144113 (2022). doi:10.1103/PhysRevB.106.144113.

[4] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics, vol. 29, Cambridge University Press, 1990.